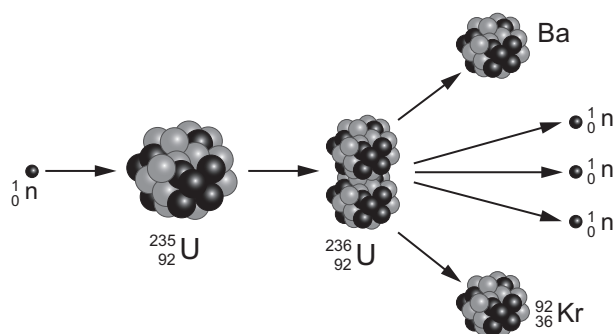
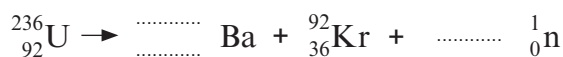


4. During their operation nuclear reactors produce large quantities of electrical energy without creating greenhouse gases.

(a) One type of reaction involving uranium-235 is shown in the diagram below.



- (i) Use the information in the diagram to complete the balanced symbol equation for part of this reaction. [3]



- (ii) Underline the correct word(s) in the brackets in the paragraph below. [4]

The absorption of slow neutrons can cause a (**fusion** / **fission** / **chemical**) reaction in uranium nuclei.

The neutrons are slowed down by (a **moderator** / **control rods** / **concrete shielding**).

The emission of (**protons** / **neutrons** / **electrons**) in this reaction can cause a chain reaction.

An uncontrolled chain reaction is prevented by using (a **moderator** / **control rods** / **concrete shielding**).

- (iii) Complete the table below about the two uranium isotopes. [2]

Uranium isotope	Number of protons	Number of neutrons
uranium-235		143
uranium-236		

- (b) Information about two natural isotopes of uranium is given below.

Isotope	Symbol	Type of decay	Half-life (years)
uranium-235	${}_{92}^{235}\text{U}$	alpha	700 million
uranium-238	${}_{92}^{238}\text{U}$	alpha	4.5 billion

- (i) Put ticks (✓) in the **two** boxes that correctly give the meaning of the half-life of a radioactive substance. [2]

Half the time for the radioactivity to halve

☐

The time taken for an atom to split in half

☐

The time taken for the number of undecayed nuclei to halve

☐

The time taken for the activity to halve

☐

The time taken for half of the alpha particles to decay

☐

- (ii) The isotopes of uranium in the table have been in existence since the Earth was formed. One estimate of the Earth's age is 4.5 billion years. Calculate the fraction of uranium-238 that remains on Earth today. [2]

Fraction =

- (iii) When the Earth was formed, it contained equal amounts of uranium-235 and uranium-238. It is suggested that more uranium-235 remains today than uranium-238. Explain whether or not you agree with this suggestion. [2]

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